Where Does Weather Matter? Isolating the Role of Weather in California Energy Markets

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Abstract

Weather is often used as an exogenous input in energy modeling, but its independent contribution has not been systematically explored. It is well understood that weather drives energy supply and demand, which in turn drive market outcomes [1]. However, existing approaches typically follow one of two paradigms: they either model market outcomes from system state variables, or use a multi-modal suite of inputs, including weather, to regress locational marginal pricing (LMP) directly [2] [3]. This study isolates and characterizes the weather-dependent signal in downstream energy market variability. We find that system state variables are weather-driven, and exhibit nonlinear temporal structure that rewards sequence-based modeling. However, forecasting market-level outcomes requires conditioning on system state rather than raw meteorological inputs.

1 Key Findings

- Weather alone has little predictive power for locational marginal pricing (LMP) and its components (congestion, energy cost).
- Calendar regimes capture most of the available signal in this setup when forecasting congestion spikes
- Net load ramp is substantially weather driven, with nonlinear temporal structure. A transformer model improves on the performance of an HGB tree at all horizons in the 1-6h range, with the performance gap widening at longer horizons.

2 Setup

California Independent System Operator (CAISO) was used as the primary data source for energy market data. From CAISO, via the gridstatus API, LMP per hub (SP15, NP15, and ZP26) was obtained, as well as regional load and fuel mix generation data, from 2023-01-01 to 2026-01-01. These were preprocessed to 1h resolution. 2023 and 2024 were used for training, and 2025 was held out for testing.

ERA5 hourly weather data was obtained from CDS, including ssrd, fdir, tcc, tp, t2m, and tcwv. The nearest ERA5 cell to each CAISO hub was identified, and a spatial average over a 3x3 ERA5 patch was taken for each variable.

In addition to the raw weather features, lag and rolling window features were computed, as well as time features which included sin/cos encodings of hour of day, day of year, and day of week.

3 Results

3.1 LMP and its components

We initially attempted to regress LMP, as well as two of its components - energy cost, and congestion, for the SP15 Southern California hub, from weather and time features. We also examined the congestion spread between Northern and Southern California. Across all targets, we found that a simple persistence baseline performed best, indicating that weather is not a direct driver of LMP. This in turn constrains where weather-based modeling can add value.

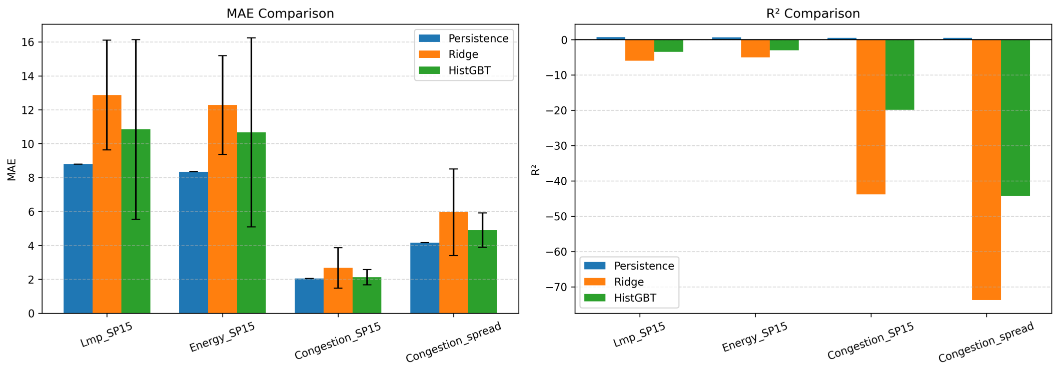

Figure 1: Direct Regression of LMP Components

3.2 Congestion Spikes

We then forecast 24-h ahead congestion spikes from time and weather features. The time features included hour of day, and month. Year and any absolute time features were excluded to prevent the model from memorizing incidents. While weather features improved the performance of the linear model, tree-based models performed better with temporal features alone. This implies that congestion spikes are calendar regime events, rather than weather-driven anomalies. We report the ROC-AUC for logistic regression and an HGB classifier across three different feature sets below.

Table 1: ROC-AUC for 24-h ahead congestion spike classification (SP15).

Model	**No Weather**	**Weather, No Lag**	**Weather + Lag**
Logistic Regression	0.731	0.777	0.801
HGB Classifier	0.793	0.789	0.789

3.3 Net load ramp forecasting

Net load represents the demand that must be met after wind and solar generation are accounted for. It is computed as load - (wind generation + solar generation). Changes, or ramping, in net load directly affect reserve needs and grid reliability. Load and generation are reported over the entire region, not per hub. Similar to [4], we forecast the change in load over 1h-6h horizons, and found that this is a case where weather drives the dependent variable. Moreover, the relationship is driven by nonlinear temporal structure in the weather signal. An HGB tree significantly outperformed both a persistence baseline and a linear regression model. In addition, we found that lag features consistently appeared in the top 20 when decomposing feature importance, which motivated the extension to a transformer approach in the next experiment.

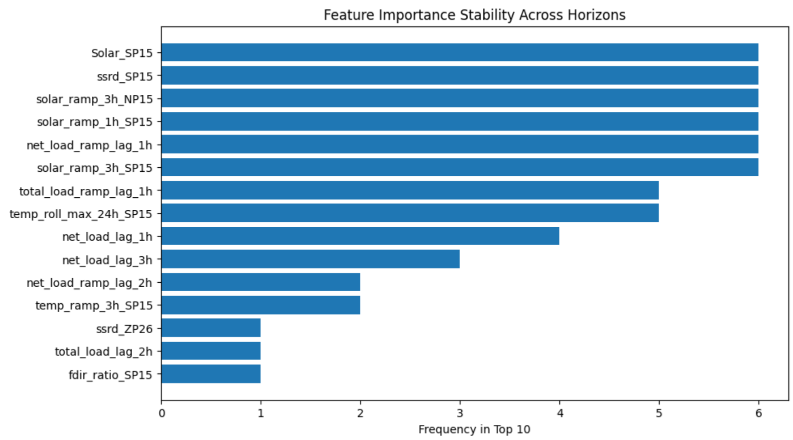

Figure 2: HGB Tree Feature Importance Across Horizons

3.4 Net load ramp temporal modeling with transformers

An encoder-only transformer with simple sin/cos positional embeddings, trained on a 168h (1 week) input window was employed to explore whether there are more complex nonlinear relationships in the time series weather data than a tree model could exploit. As there were only three hubs in this experiment, there was little spatial structure to exploit, so the experiment focused solely on temporal signal. The transformer outperformed the HGB tree at all horizons, but the performance gap widened noticeably as horizon length increased. This suggests that complex, nonlinear long-range dependencies are present, and are not sufficiently captured by lag features alone. We report MAE in GW across 6 horizons below for the full suite of models.

Table 2: Net load ramp MAE (GW) by forecast horizon.

Model	**1h**	**2h**	**3h**	**4h**	**5h**	**6h**
Persistence	1.491	3.106	4.856	6.450	7.996	9.317
Linear Regression	1.153	2.171	2.886	3.431	3.827	4.045
HGB	0.512	0.980	1.368	1.650	1.939	2.153
Transformer	0.455	0.758	0.966	1.146	1.298	1.458

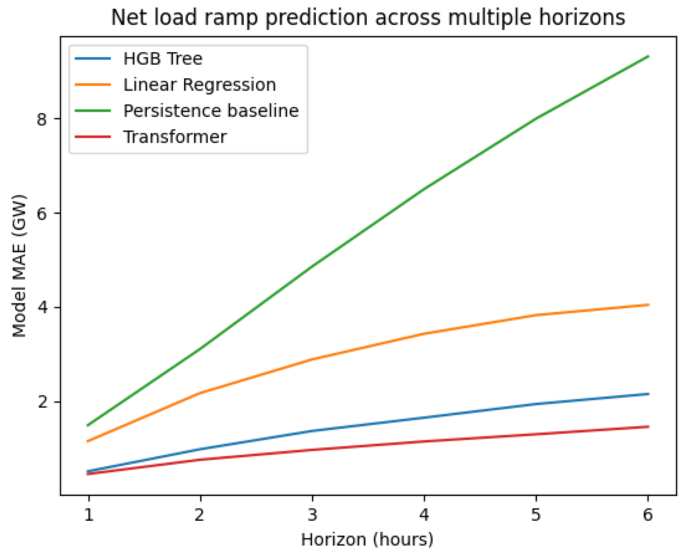

Figure 3: Comparison of Models Across Horizons

4 Interpretation

The results reveal a clear asymmetry between the role of weather in the physical and market layers of the power system. Market outcomes in the day-ahead setting (LMP, congestion, energy cost) are dominated by temporal and institutional structure - transmission limits, bidding structure, grid topology, etc - and weather adds little useful signal. The physical layer (load, net load ramp) is weather-driven, and the relationship is defined by nonlinear temporal dynamics. This suggests that weather is a useful input for forecasting system state, but not a sufficient signal for down- stream market outcomes, which rely on context beyond the physical layer. In this study we utilized only historical weather reanalysis data, to avoid the uncertainty introduced by forecast weather, but from this, we can hypothesize that downstream models incorporating weather forecasts might be better served by conditioning on weather-forecast-driven system state rather than raw weather forecast data.

5 Future Directions

- Learn a structured intermediate state representation which feeds into downstream market forecasting models. Weather does not fully determine the intermediate system state, but when coupled with grid constraints such as outages, transmission limits etc, a learned latent representation could provide a spatially dense and far richer input to downstream market layer forecasting models than load or generation, which are spatially coarse and incomplete projections of the drivers of market dynamics.
- Incorporate Weather forecasts - this study intentionally restricted the weather inputs to historical reanalysis data to avoid the uncertainty introduced by forecast weather. In practice, however, day-ahead market forecasting typically relies on forecast weather data. A natural extension of this study would be to quantify introduced uncertainty from weather forecasts, derive forecast system state from weather forecasts and compare effectiveness of conditioning on weather forecasts vs weather-driven system state forecasts.
- Multi-modal inputs and outputs - expanding to more than just weather data, as weather is not the primary driver in a lot of these cases. Deep learning makes more sense as the complexity of interactions increases, both in the input and output layers.

References

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